



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

MATHEMATICS P1

MARKING GUIDELINE

MARKS: 100

This marking guideline consists of 6 pages.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept any correct alternative method. Give follow-through marks (CA) where a learner uses their own earlier wrong answer correctly. Answers only are not awarded full marks where working is required.

QUESTION 1

$$\begin{aligned} 1.1.1 \quad &= 2x^3 + 8x^2 - 2x - 3x^2 - 12x + 3 \quad \checkmark \checkmark \quad (3) \\ &= 2x^3 + 5x^2 - 14x + 3 \quad \checkmark \end{aligned}$$

$$\begin{aligned} 1.1.2 \quad &= \frac{(x-3)(x+3)}{(x+3)(x-2)} \times \frac{x-2}{x-3} \quad \checkmark \checkmark \checkmark \text{ (factorising, multiplying by the reciprocal)} \quad (4) \\ &= 1 \quad \checkmark \end{aligned}$$

$$1.2.1 \quad = 3(x^2 - 4) \quad \checkmark = 3(x-2)(x+2) \quad \checkmark \quad (2)$$

$$1.2.2 \quad = (2x+3)(x-5) \quad \checkmark \checkmark \quad (2)$$

$$1.2.3 \quad = (x-2) \quad \checkmark (x^2 + 2x + 4) \quad \checkmark \quad (2)$$

$$1.2.4 \quad = a(x-y) + 2(x-y) \quad \checkmark = (x-y)(a+2) \quad \checkmark \quad (2)$$

$$1.3.1 \quad = \frac{2^x(2^2-1)}{3 \cdot 2^x} \quad \checkmark = \frac{2^x \cdot 3}{3 \cdot 2^x} \quad \checkmark = 1 \quad \checkmark \quad (3)$$

$$1.3.2 \quad = (3^3)^{2/3} \times (2^2)^{-1/2} \quad \checkmark = 9 \times \frac{1}{2} \quad \checkmark = \frac{9}{2} \text{ (or 4,5)} \quad \checkmark \quad (3)$$

$$1.3.3 \quad = x^6 y^{-4} \times x^{-4} y^5 \quad \checkmark = x^{6-4} y^{-4+5} \quad \checkmark = x^2 y \quad \checkmark \quad (3)$$

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QUESTION 2

$$2.1.1 \quad 3x - 6 = 2x + 5 \quad \checkmark \therefore x = 11 \quad \checkmark \quad (2)$$

$$2.1.2 \quad (x-7)(x+2) = 0 \quad \checkmark \therefore x = 7 \quad \checkmark \text{ or } x = -2 \quad \checkmark \quad (3)$$

$$\begin{aligned} 2.1.3 \quad &3(x+1) - 2(x-3) = 12 \quad \checkmark \text{ (multiply by LCD 6)} \quad (3) \\ &3x + 3 - 2x + 6 = 12 \quad \checkmark \therefore x = 3 \quad \checkmark \end{aligned}$$

$$2.1.4 \quad 3^{x+1} = 3^4 \quad \checkmark \therefore x+1 = 4, x = 3 \quad \checkmark \quad (2)$$

$$2.1.5 \quad x^2 - 3x - 10 = 0 \quad \checkmark (x-5)(x+2) = 0 \therefore x = 5 \quad \checkmark \text{ or } x = -2 \quad \checkmark \quad (3)$$

$$2.2 \quad -2 < 2x \leq 8 \quad \checkmark \therefore -1 < x \leq 4 \quad \checkmark \quad (3)$$

Number line: open circle at -1, closed (shaded) circle at 4, line between them ✓

$$\begin{aligned} 2.3 \quad &\text{From } 3x - y = 7: y = 3x - 7 \quad \checkmark \quad (4) \\ &x + 2(3x - 7) = 7 \quad \checkmark \rightarrow 7x = 21 \therefore x = 3 \quad \checkmark; y = 3(3) - 7 = 2 \quad \checkmark \end{aligned}$$

$$2.4 \quad A - P = Pin \checkmark \therefore i = \frac{A - P}{Pn} \checkmark \quad (2)$$

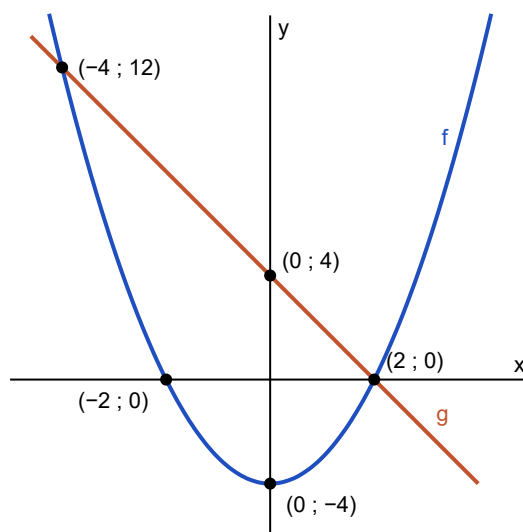
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QUESTION 3

- 3.1 $21 \checkmark$; $25 \checkmark$ (2)
- 3.2 Common difference $d = 4 \checkmark$; $T_n = 4n + 1 \checkmark$ (2)
- 3.3 $T_{50} = 4(50) + 1 \checkmark = 201 \checkmark$ (2)
- 3.4 $4n + 1 = 157 \checkmark \therefore n = 39$, it is the 39th term $\checkmark$ (2)
- 3.5 $4n + 1 = 100 \rightarrow n = 24,75 \checkmark$ (2)
 n must be a natural number, so 100 is not a term $\checkmark$

[10]**QUESTION 4**

- 4.1 $(0 ; -4) \checkmark$ (1)
- 4.2 $x^2 - 4 = 0 \rightarrow x = \pm 2 \checkmark$; $(-2 ; 0)$ and $(2 ; 0) \checkmark$ (2)
- 4.3 $-2x + 4 = 0 \rightarrow (2 ; 0) \checkmark$ (1)
- 4.4 (5)



Marks: shape of $f \checkmark$; turning point $(0 ; -4) \checkmark$; x-intercepts of $f \checkmark$; straight line g with negative gradient $\checkmark$; intercepts of g : $(0 ; 4)$ and $(2 ; 0) \checkmark$

- 4.5 $y \geq -4$ (or $y \in [-4 ; \infty)$) $\checkmark$ (1)

4.6 $x^2 - 4 = -2x + 4 \rightarrow x^2 + 2x - 8 = 0 \checkmark$ (4)
 $(x + 4)(x - 2) = 0 \rightarrow x = -4 \text{ or } x = 2 \checkmark$
 $f(x) \geq g(x) \text{ when } x \leq -4 \checkmark \text{ or } x \geq 2 \checkmark$

4.7 $q = 1 \checkmark$ (3)
 $4 = \frac{a}{2} + 1 \checkmark \therefore a = 6 \checkmark$

4.8.1 $y = 0 \checkmark$ (1)

4.8.2 $(0 ; 1) \checkmark$ (1)

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QUESTION 5

5.1 $A = P(1 + in) = 8\,000(1 + 0,07 \times 5) \checkmark\checkmark = R10\,800 \checkmark$ (3)

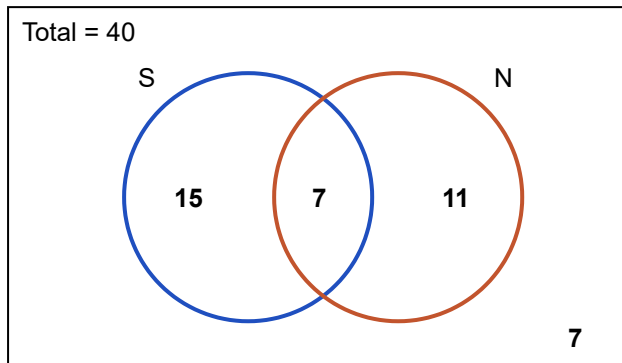
5.2 $A = P(1 + i)^n = 8\,000(1,065)^5 \checkmark\checkmark = R10\,960,69 \checkmark$ (3)

5.3 Sipho $\checkmark$, by R160,69 $\checkmark$ (2)

5.4 $A = 6\,500(1,05)^3 \checkmark\checkmark = R7\,524,56 \checkmark$ (3)

5.5 $\frac{2\,300}{18,40} \checkmark = \$125 \checkmark$ (2)

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QUESTION 6**6.1****(4)**

Marks: both circles inside the rectangle ✓; 7 in the overlap ✓; 15 (only S) and 11 (only N) ✓; 7 outside both circles ✓

6.2.1

$$P(\text{neither}) = \frac{7}{40} \checkmark \checkmark$$

(2)**6.2.2**

$$P(S \text{ or } N) = \frac{15 + 7 + 11}{40} \checkmark = \frac{33}{40} \checkmark$$

(2)**6.2.3**

$$P(\text{only } S) = \frac{15}{40} \checkmark = \frac{3}{8} \checkmark$$

(2)**6.3**

No ✓. 7 learners play both, so S and N can happen together ($n(S \text{ and } N) \neq 0$) ✓

(2)**[12]****TOTAL: 100****END OF MARKING GUIDELINE**